Student Exam Performance Analysis

Midterm Assessment

DATA 581 - Data Visualization | Gonzaga University

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Purpose

I was asked to analyze exam results for 1,000 students and explain what separates higher performers from lower ones. The aim is to give decision makers a clear picture of where extra support would help most. I looked at three subjects alongside background details such as gender, parental education, whether a student completed a test preparation course, and which lunch program they were on.

What I Found

Three patterns stand out. First, preparation pays off. Students who completed the test preparation course scored higher in every subject, about 7.6 points higher on average. This is encouraging, because it is something a school can directly influence. Second, economic background appears to matter even more. Students on the standard lunch program scored about 8.6 points higher than students on the free or reduced lunch program, the largest gap I found. Lunch status is commonly used as an indicator of household income, so this gap likely reflects financial circumstances at home rather than anything about the student's ability. The gap was widest in math. Third, reading and writing go hand in hand. A student who reads well almost always writes well; the two scores rise and fall together almost perfectly. Math is related too, but a little more independent. Gender differences were smaller and depended on the subject: girls scored higher in reading and writing, while boys scored slightly higher in math.

Recommendations

- Expand access to the test preparation course, since it helped across every subject and is within the school's control.
- Direct additional academic support toward students on the free or reduced lunch program, where the largest gap appears, recognizing that some of this support may need to address needs outside the classroom.

- Treat reading and writing as a single literacy focus, because improving one tends to lift the other, which makes support more efficient.

Data Overview

The dataset has no missing values. Columns were renamed to remove spaces and slashes for cleaner code, and a single `average_score` column was created to summarize each student with one number.

Table 1: Dataset Variables

Variable	**Description**
gender	Female or male
race_ethnicity	Anonymized group label (A through E)
parental_education	Highest education level of a parent
lunch	Standard or free/reduced (a common stand-in for household income)
test_prep	Completed or none (our stand-in for study habits)
math score	Exam result, 0–100
reading score	Exam result, 0–100
writing score	Exam result, 0–100
average_score	Mean of math, reading, and writing (derived)

Summary Statistics

Average scores sit in the high 60s. Medians are almost identical to the means, so the scores are fairly symmetric rather than skewed. Writing and math show the widest spread, while the combined average is the most stable because averaging three subjects smooths out individual swings.

Table 2: Descriptive Statistics by Subject

Subject	**Mean**	**Median**	**Std Dev**	**Variance**
Math	66.09	66.0	15.16	229.92
Reading	69.17	70.0	14.60	213.17
Writing	68.05	69.0	15.20	230.91
Average Score	67.77	68.3	14.26	203.27

Exploratory Data Analysis

Chart 1: Average Score by Test Preparation

Does completing the test preparation course relate to higher scores?

Students who completed the preparation course average about 72.7, versus 65.0 for those who did not, a gap of roughly 7.6 points. The gain holds in every subject and is largest in writing. The y-axis starts at zero so the bar heights represent the scores honestly.

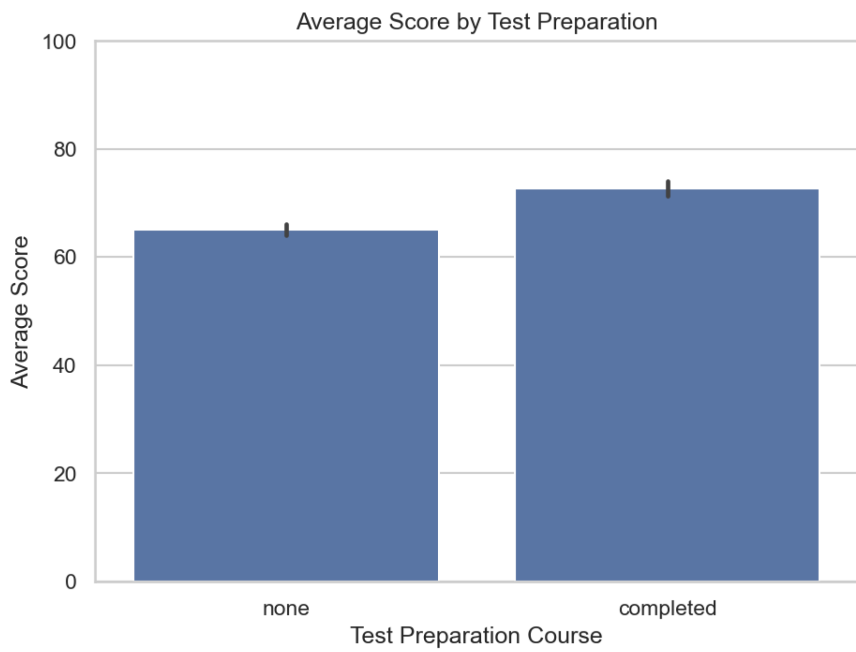

Figure 1: Average score by test preparation course completion.

Chart 2: Distribution of Average Scores

How are overall scores spread across all students?

The distribution is roughly bell-shaped and centered near 68, with most students between about 55 and 80. A thin left tail shows a small number of low performers, which is where targeted support would matter most.

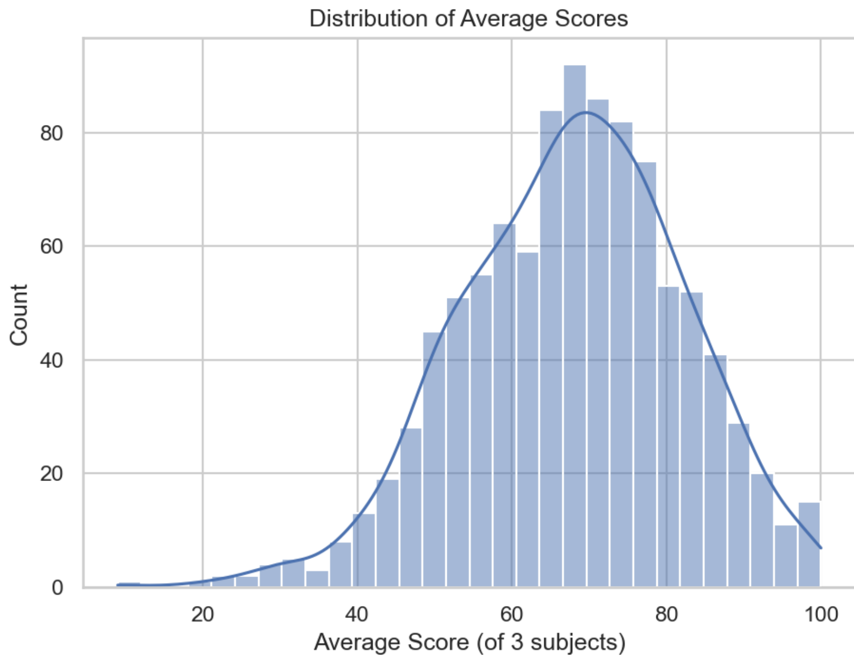

Figure 2: Distribution of average scores across 1,000 students with KDE overlay.

Chart 3: Reading vs. Writing Score Relationship

Do reading and writing scores move together?

Reading and writing are very strongly correlated, so a student strong in one is almost always strong in the other. Math is also strongly related to both but a little more independent. In practice, reading and writing partly measure the same underlying literacy skill.

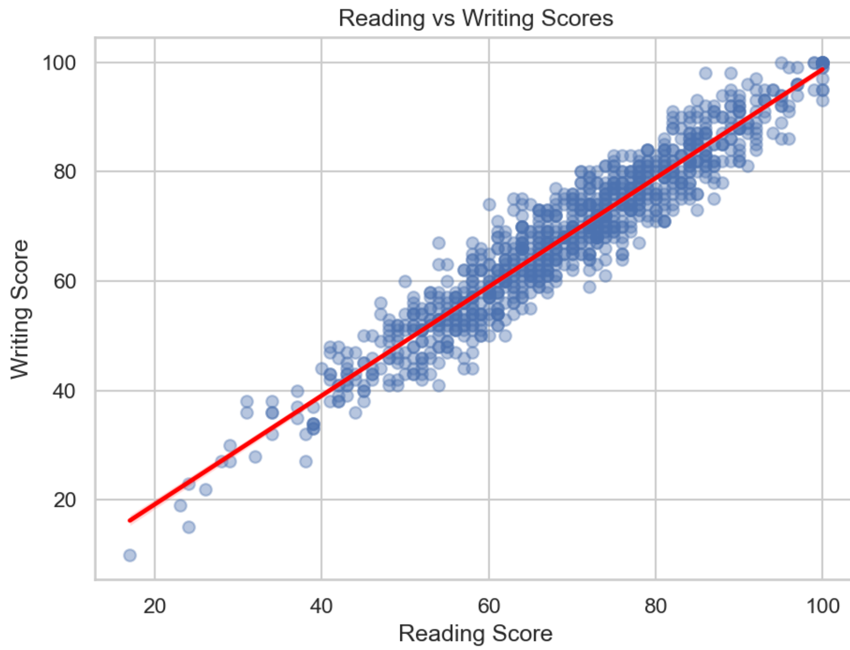

Figure 3: Scatter plot of reading vs. writing scores with regression line.

Table 3: Correlation Matrix

	Math	**Reading**	**Writing**
Math	1.000	0.818	0.803
Reading	0.818	1.000	0.955
Writing	0.803	0.955	1.000

Chart 4: Score by Lunch Program

Students on the standard lunch average about 70.8, versus 62.2 for those on free or reduced lunch, a gap of roughly 8.6 points, the largest of any factor here. The gap is widest in math (about 11 points). Because lunch status reflects economic background, this points to support being needed beyond the classroom.

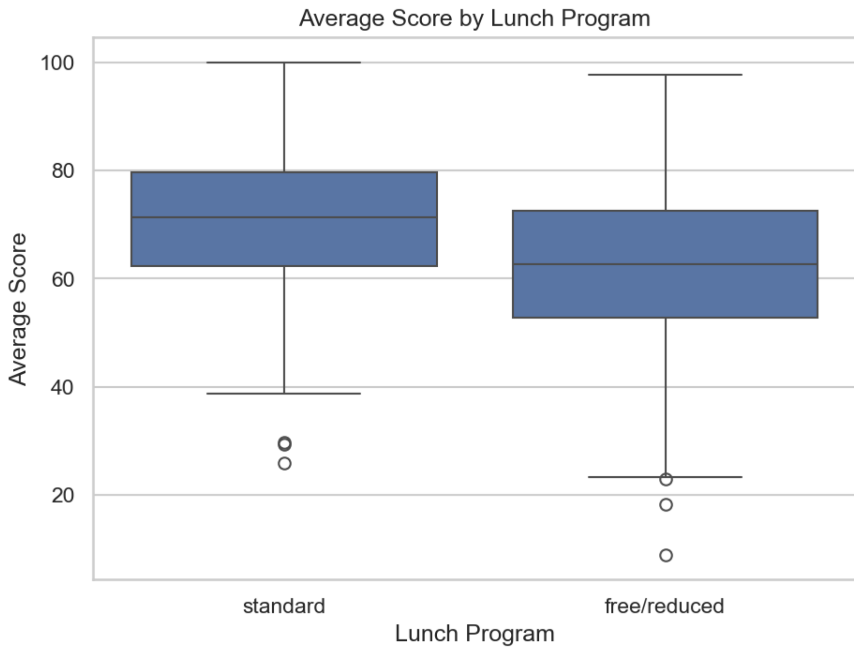

Figure 4: Boxplot of average scores by lunch program status.

Technical Report

Introduction

Schools want to understand which factors are associated with stronger exam performance so they can target limited support effectively. This report analyzes a dataset of 1,000 students, each with scores in three subjects (math, reading, and writing) and several background attributes: gender, race/ethnicity (anonymized), parental level of education, lunch program status, and whether the student completed a test preparation course. The question I set out to answer is which of these factors relate most clearly to performance, and what a school might do in response.

Methods

Data preparation. I loaded the dataset and confirmed it contained no missing values, so no imputation was needed. I renamed the columns to remove spaces and slashes for cleaner code, and I created an `average_score` column equal to the mean of the three subject scores, giving one summary measure per student.

Analysis approach. I computed summary statistics (mean, median, and variance) for each subject and for the combined average. I then examined the data through four chart types:

a categorical comparison (average score by test preparation), a distribution (the spread of average scores), a relationship (reading versus writing), and a correlation view, plus an additional chart for gender. All work was done in Python using pandas, seaborn, and matplotlib.

Results

Summary statistics show average scores in the high 60s (math 66.1, reading 69.2, writing 68.1) with medians almost equal to the means, showing roughly symmetric distributions. The categorical comparison (Chart 1) shows students who completed the test preparation course averaging 72.7 against 65.0 for those who did not, a difference of about 7.6 points that holds across all three subjects. The distribution (Chart 2) is approximately bell-shaped and centered near 68, with a thin left tail of low performers. The relationship chart (Chart 3) and the correlation table show reading and writing are very strongly correlated ($r = 0.96$), while math correlates with each at about 0.80. The lunch comparison (Chart 4) reveals the largest gap in the data: standard-lunch students average 70.8 versus 62.2 for free or reduced lunch, a difference of about 8.6 points, widest in math.

Discussion

Two factors stand out as most associated with performance: lunch program status and test preparation. Because lunch status is a standard proxy for household income, its large effect suggests that economic circumstances outside school weigh heavily on results, particularly in math. Test preparation, by contrast, is something the school controls directly, and its consistent benefit across subjects makes it an appealing lever. The very high correlation between reading and writing implies the two are partly measuring the same underlying literacy skill, so they can be supported together. Throughout, it is important to stress that these are observational associations. The data cannot establish cause, and unmeasured factors (school quality, home environment, prior achievement) may drive several of these patterns at once.

Conclusion

The analysis points to three actions: expand the test preparation course, since it helps every subject and lies within the school's control; focus additional support on students on the free or reduced lunch program, where the largest gap appears; and treat reading and writing as a combined literacy effort. These recommendations should be read as informed starting points for targeted follow-up, not as proven cause-and-effect relationships.

Ethics Reflection

Visualizations can mislead an audience even when no one intends to deceive. The most common way is the axis: starting a bar chart above zero makes small differences look dramatic. Choosing a narrow range, cropping a time series, or stretching one axis can manufacture a trend that is not there. Aggregation is another quiet source of error. Combining everyone into a single average can hide important subgroups, and a pattern that holds for the whole group can even reverse within a subgroup. Color can imply meaning that does not exist, and common red-green palettes exclude colorblind readers. Leaving out sample sizes or any sense of uncertainty makes a fragile result look solid, and placing a strong title next to a correlation nudges viewers to assume one thing caused another.

Because of all this, analysts carry a clear responsibility when they present results. First, encode data honestly: use proportional baselines so that visual size matches actual magnitude. Second, provide context, including how many observations there are, what time period is covered, and what a variable actually measures. In this project, for example, lunch status is a stand-in for income, not a judgment about a student, and saying so plainly matters. Third, separate evidence from recommendation, and label associations as associations rather than dressing them up as proof. Fourth, design for the actual audience: a non-technical reader will take a chart at face value, so the burden of accuracy sits with the analyst, not the viewer. Finally, disclose limitations openly instead of hoping no one asks. The underlying principle is simple: since the audience usually cannot inspect the data or the code behind a chart, the analyst's job is to make the honest reading the easiest one to reach.

Dataset source: Students Performance in Exams, Kaggle.